Progress Report on Studies of Basic Discretization and Linearization Methods to Realize System State Predictions and System Control

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1 Introduction

Computer simulations are extensively employed in various fields to provide numerical predictions of system states. They can be used to study, test, design, and optimize complex systems or phenomena that are difficult or expensive to experiment with in reality. An example of the applications of computer simulations is weather forecasting, where computer simulations are used to build models based on the physical laws governing atmospheric dynamics. Current weather observations are used as initial conditions for the simulation models to solve numerical equations to project future changes in different weather parameters, such as temperature, pressure, wind, humidity, and precipitation [1]. Computer simulations can also be applied in the field of pandemic prevention, where they can be used to study and predict the spread of infectious diseases such as COVID-19. The outcomes of a pandemic could be simulated by taking the transmission rate, the reproduction number, the peak incidence, and the number of total cases as system state variables [2].

Natural physical systems can be considered as continuous-time systems whose system states could be expressed by time functions. Besides, most natural systems could only be described using non-linear models. To perform digital computations for natural physical systems, it is necessary to transform the original non-linear continuous-time systems into their linear discrete-time form. This is due to the digital properties of transistors, the computational units used in computers. This transformation involves the application of discretization methods and linearization methods.

An example of a conventional discretization method is the forward-shift method, which discretizes a linear continuous state space model using the forward-shift operator. The most widely used discretization method is the forward-difference method, which adapts the same form and parameters as those of continuous-time systems and converts the continuous-time differential operator into its discrete-time equivalent, the delta operator [3]. The nonlinearities of the original systems could be compensated by using a variety of linearization methods, such as linearization by change of states and linear approximation [4].

2 Conversion from continuous-time models to discrete-time models

To achieve the interaction of a physical system with a computer, we use an interface consisting of an analog-to-digital (A/D) converter and a digital-to-analog (D/A) converter. Here, the A/D converter achieves analog-to-digital conversion by sampling the continuous-time signal $y(\cdot)$ with a fixed sampling interval Δ to obtain the sample values $y(k\Delta)$, where k is a non-negative integer [4]. We decide the type of discretization method depending on whether the system is described by a general nonlinear state space model or a linear state space model. In this section, we will introduce the basics of discretization methods.

2.1 General systems

For general systems that nonlinear models describe, it is not possible to obtain an exact solution to the underlying continuous-time equations. However, we can approximate the solutions by using numerical methods such as the Runge-Kutta method or predictor corrector formulas. Among these methods, the Euler approximation has the highest simplicity. Let us consider an example of a finite-dimensional system that is described by equations

$$\frac{d}{dt}x(t) = f(x(t), u(t), d(t), t), \qquad x(t_0) = x_0 \quad (1)$$

and

$$y(t) = g(x(t), u(t), d(t), t)$$

(2)

where the system state is denoted by x(t), the input to the system by u(t), the disturbance by d(t), the output by y(t), the starting time by t_0, and the initial state by $x(t_0)$. For such a system, we can approximate the solution to equations (1) and (2) as

$$x[(k+1)\Delta] = x[k\Delta] + \Delta f[x(k\Delta), u(k\Delta), d(k\Delta), k\Delta] \quad (3)$$

and

$$y(k\Delta) = g[x(k\Delta), u(k\Delta), d(k\Delta), k\Delta] \quad (4)$$

2.2 Systems with linear state space models

For physical systems with linear state space models, it is possible to obtain an exact solution to the underlying continuous-time equations and, therefore an exact description of the sampled response. If we consider a continuous time linear state space model shown by equations

$$\frac{d}{dt}x(t) = Ax(t) + Bu(t) \quad (5)$$

$$y(t) = Cx(t) \quad (6)$$

and assume that the system is in state x(kΔ) at the time $t = k\Delta$, we can compute the response for $t \geq k\Delta$. The exact solution of equation (5) is given by

$$x(t) = e^{A(t-k\Delta)}x(k\Delta) + \int_{k\Delta}^{t} e^{A(t-\tau)}Bu(\tau)d\tau \quad (7)$$

where e^A stands for the matrix power series:

$$e^{At} \triangleq I + At + \frac{1}{2!}(At)^2 + \frac{1}{3!}(At)^2 + \cdots \quad (8)$$

If we define the input as a zero-order hold, which means we have

$$u(t) = u(k\Delta), for \ \ k\Delta \leq t < (k+1)\Delta \quad (9)$$

Substituting (9) into (7) gives us

$$x\big((k+1)\Delta\big) = e^{A\Delta}x(k\Delta) + \left[\int_0^{\Delta} e^{A(\Delta-\tau)}Bd\tau\right]u(k\Delta) \quad (10)$$

Therefore, we can obtain a discrete time model that describes this system expressed as the following equations:

$$x_{k+1} = Fx_k + Gu_k \quad (11)$$

$$y_k = Hx_k \quad (12)$$

where

$$x_k \triangleq x(k\Delta) \quad (13)$$

$$u_k \triangleq u(k\Delta) \quad (14)$$

$$y_k \triangleq y(k\Delta) \quad (15)$$

$$F \triangleq e^{A\Delta} \quad (16)$$

$$G = \int_0^{\Delta} e^{A(\Delta-\tau)}Bd\tau = A^{-1}(e^{A\Delta} - I)B \quad (17)$$

$$H = C \quad (18)$$

Among these, equation (17) is only valid when A is nonsingular [4].

2.3 The shift operator

Since the original linear continuous time model is expressed using an operator d/dt, it is convenient to also use operator notation to express the discrete time form of the model. One operator that describes discrete time models is the forward shift operator q, with a definition

$$qx_k \triangleq x_{k+1} \quad (19)$$

If we substitute equation (19) into equation (11), we can obtain a more compact form of the discrete time model:

$$qx_k = Fx_k + Gu_k \quad (20)$$

$$y_k = Hx_k \quad (21)$$

However, since the operator q is far different from the continuous time differential operator d/dt, the obtained discrete time model has a low correspondence to the original continuous time model, leading to low resemblance of future system states[4].

2.4 The delta operator

We can improve the operator notation form of discrete time model by replacing the shift operator with the delta operator, δ, that functions similarly to a derivative. The delta operator is defined as

$$\delta \triangleq \frac{q-1}{\Delta} \quad (22)$$

Substituting equation (22) into equation (11) gives us

$$\delta x_k = \frac{x_{k+1}-x_k}{\Delta} = \frac{x(k\Delta+\Delta)-x(k\Delta)}{\Delta} \quad (23)$$

Here, since the delta operator is related with the shift operator by a simple linear equation as shown in

(22), it is as flexible as the shift operator in describing a discrete time system. However, it offers a higher resemblance with the original continuous time system since it is similar to the continuous time differential operator d/dt [4].

3 Conversion from nonlinear state space models to linear state space models

Systems with linear state space models are generally easier to analyze than those with nonlinear models, as nonlinear systems exhibit more complex behaviors. [4]. Therefore, a complicated system requires linearization, converting nonlinear state space models into linear ones. In this section, we will introduce the basics of linearization methods.

3.1 Linearization by change of states

Let us consider a general state space model of the form

$$\frac{d}{dt}x = f(x,u) \tag{24}$$

When the output is $y = \gamma(x)$ so that y and first n-1 derivatives of y with respect to time are independent of the input u, then we can define a new state vector as

$$\bar{x}^T \triangleq (y, \dot{y}, \cdots, y^{n-1}) \tag{25}$$

Using this new state vector, we can transform the original model shown by equation (24) into

$$\left.\begin{aligned} &\frac{d}{dt}\bar{x}_1 = \bar{x}_2 \\ &\qquad\vdots \\ &\frac{d}{dt}\bar{x}_{n-1} = \bar{x}_n \end{aligned}\right\} \tag{26}$$

$$\frac{d}{dt}\bar{x}_n = y^n \triangleq \theta(\bar{x},u)$$

Supposing the function $\theta(\bar{x}, u)$ is invertible with respect to u, then we can solve $\theta(\bar{x},u) = u'$ for u in terms of u' and $\bar{x}$, leading to the following state equations:

$$\left.\begin{aligned} &\frac{d}{dt}\bar{x}_1 = \bar{x}_2 \\ &\qquad\vdots \\ &\frac{d}{dt}\bar{x}_{n-1} = \bar{x}_n \end{aligned}\right\} \tag{27}$$

$$\frac{d}{dt}\bar{x}_n = u'$$

which describe a linear system [4].

3.2 Linear approximation

We can describe a nonlinear system that operates with small deviations about an operating point by the following set of equations

$$x(t) = X + \Delta x(t) \tag{28}$$

$$u(t) = U + \Delta u(t) \tag{29}$$

$$y(t) = Y + \Delta y(t) \tag{30}$$

where the three constants X,Y, and Z define the operating point of the system, and ΔX, ΔY, and ΔZ define the small deviations of the system state, the input to the system, and the system output respectively. If we assume that at the equilibrium point, we have

$$f(x,u) = 0 \tag{30}$$

$$Y = g(x,u) \tag{31}$$

For such a system, we can obtain an approximate model using Taylor's series approximation, as shown by the following equations:

$$\frac{d}{dt}\Delta x(t) = \frac{d}{dt}x(t) = f\big(X + \Delta x(t), U + \Delta u(t)\big) \approx \left(\frac{\partial f}{\partial x}\right)\Big|_{x=X,u=U}\Delta x(t) + \left(\frac{\partial f}{\partial u}\right)\Big|_{x=X,u=U}\Delta u(t) \tag{32}$$

$$\Delta y(t) = g\big(X + \Delta x(t), U + \Delta u(t)\big) - Y \approx \left(\frac{\partial g}{\partial x}\right)\Big|_{x=X,u=U}\Delta x(t) + \left(\frac{\partial g}{\partial u}\right)\Big|_{x=X,u=U}\Delta u(t) \tag{33}$$

This linearization technique involves differentiating vector functions f(x) and g(x) with respect to vectors x and u, hence computation of Jacobian matrices is also required. Subsequently, we can let $A = \left(\frac{\partial f}{\partial x}\right)\Big|_{x=X,u=U}$, where the differential $\partial f/\partial x$ will give n×n matrices with i, jth elements expressed by $\partial f_i/\partial x_j$. Similarly, we can let B, C, and D represent the three other vector function differentials. Therefore, the original system can have the standard linear state space form [4]:

$$\frac{d}{dt}(\Delta x) \approx A(\Delta x) + B(\Delta u) \tag{34}$$

$$\Delta y \approx C(\Delta x) + D(\Delta u) \tag{35}$$

4 Future prospects

In PBL I, we studied the basics of traditional ways to convert nonlinear continuous time models to linear discrete time models to enhance the analyzability of natural systems. In the future, we plan to investigate the fundamental ways of predicting the system state behaviors with simulations using linear discrete time models. We also aim to conduct research on controlling and optimizing the state responses of physical plants. Our initial goal is to gain a comprehensive understanding of controlling techniques for feedback systems.

References

[1] US Department of Commerce, NOAA. “The Forecast Process - Forecasting the Future.” *National Weather Service*, NOAA’s National Weather Service, 16 Aug. 2015, www.weather.gov/rah/virtualtourforecast, accessed on 13 Feb. 2024

[2] Adam, David. “Special Report: The Simulations Driving the World’s Response to Covid-19.” *Nature News*, Nature Publishing Group, 2 Apr. 2020, www.nature.com/articles/d41586-020-01003-6, accessed on 14 Feb. 2024

[3] Nguyen-Van, Triet, and Noriyuki Hori. “New class of discrete-time models for non-linear systems through discretisation of integration gains.” *IET Control Theory & Applications*, vol. 7, no. 1, Jan. 2013, pp. 80–89, https://doi.org/10.1049/iet-cta.2012.0010.

[4] Middleton, Richard H., and Graham Clifford Goodwin. *Digital Control and Estimation: A Unified Approach*. Prentice Hall, pp. 7-50, 1990.